

Performance Evaluation

PPO × Macro Adaptive Strategy (Live Trading)

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Outline

1. Performance Analysis
2. Cost Analysis
3. Backtest vs. Live
4. Strategy Monitoring
5. Data Reconciliation
6. Summary & Next Steps

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Live Trading Performance Metrics

- Same core metrics as backtest analysis, now computed on **live realised returns**:
 - ▶ CAGR (Compound Annual Growth Rate)
 - ▶ Annualised Volatility
 - ▶ Sharpe Ratio (risk-free rate = 0%, or SOFR)
 - ▶ Maximum Drawdown (peak-to-trough)
 - ▶ Beta relative to SPY (should remain near target ≈ 0.37)
 - ▶ Alpha (excess return over CAPM)
- Strategy-specific metrics (especially for PPO layer):
 - ▶ Win rate (proportion of profitable positions)
 - ▶ Average hold period per trade
 - ▶ Turnover (daily / monthly)
 - ▶ Gross leverage utilisation

[Placeholder: Live vs. Backtest Metrics Table]
(CAGR, Sharpe, Max DD, Win Rate, Turnover, etc.)

Rolling Performance Windows

- **Rolling 3-month / 6-month / 12-month** Sharpe and drawdowns are tracked daily.
- Rolling volatility compared to long-term average (backtest + live).

[Placeholder: Rolling Sharpe & Volatility Chart]
(3-month rolling vs. historical distribution)

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Live Cost Breakdown

- Every trade incurs explicit and implicit costs.
- Our live cost tracking separates:

Cost Component	Realised Live (bps of NAV)
Commissions (IB tiered)	[Placeholder]
Slippage (0.5 bps/side)	[Placeholder]
Borrow cost (3% p.a. on shorts)	[Placeholder]
Dividend withholding tax (30%)	[Placeholder]
Total annualised cost	[Placeholder]

- Compare against backtest cost assumptions.
- Identify any unexpected expenses (e.g., special borrow rates, exchange fees).

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Attribution of Differences

Potential sources:

- Entry / exit signals (data mismatch)
- Security selection (universe changes)
- Executed prices (slippage / liquidity)
- Position weights (rounding / beta-neutral adjustment)

Operational factors:

- Data source delays
- Market impact (large orders)
- Timing differences (scheduled vs. actual)
- Lack of inventory to short
- Illiquidity (wide spreads)
- Code bugs / mishandling

[Placeholder: Bar chart of P&L attribution]
(Backtest P&L minus Live P&L by category)

Focus on Most Negative Impacts

- Rank all differences by **impact on profitability**.
- Top 1–2 issues will be addressed with concrete changes:

Observed Issue	Proposed Mitigation
[Example: higher slip-page on small caps]	[Example: filter universe by avg dollar volume]
[Placeholder]	[Placeholder]
[Placeholder]	[Placeholder]

- Adjust strategy logic for real-world constraints.
- Refine execution algorithms (VWAP, chunking).
- Update cost models with live data feedback.

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Tracking Strategy Health

- **Rolling performance** window (e.g., 3 months) compared to backtest window:
 - ▶ If rolling Sharpe drops below 0.3 (or $0.5 \times$ backtest Sharpe) $\rightarrow$ alert.
 - ▶ If rolling volatility exceeds backtest 95th percentile $\rightarrow$ review.
- **Drawdown monitoring:**
 - ▶ Current drawdown vs. historical max drawdown (backtest + live).
 - ▶ Threshold: *exceed historical max by more than X%* $\rightarrow$ kill switch.

[Placeholder: Drawdown Evolution Chart]
(Current drawdown vs. historical maximum)

Kill Switch & Capacity Limits

- **Kill switch conditions (automated):**

1. Current drawdown $>$ Historical Max Drawdown $\times 1.25$
2. Daily loss $>$ 10% (circuit breaker)
3. Gross leverage $>$ $1.5\times$ for more than 2 consecutive days

Action: liquidate all positions within 1 day, switch to cash, send email alert.

- **Capacity limits:**

- ▶ Monitor AUM relative to average daily volume (ADV) of held stocks.
- ▶ If our average position $>$ 0.5% of ADV $\rightarrow$ start decomposing into child orders.
- ▶ Hard cap: 2% of any stock's ADV.

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Broker vs. Internal Records

- Daily comparison between QuantConnect records and Interactive Brokers statements.
- Key checks:

Item	Reconciliation Status
Open positions list	[Match / Mismatch]
Realised P&L (per trade)	[Match / Mismatch]
Commissions & fees	[Match / Mismatch]
Current NAV (liquidation value)	[Match / Mismatch]
Dividends received	[Match / Mismatch]
Borrow fees charged	[Match / Mismatch]

- Any discrepancy $> 0.1\%$ of NAV triggers manual review.
- Persistent errors indicate potential broker feed issues or internal calculation bugs.

Reconciliation Workflow

1. Export daily broker report (CSV/XML) after market close.
2. Run automated reconciliation script.
3. Highlight unexpected costs or missing trades.
4. If discrepancy $>$ threshold, pause next day's trading until resolved.

[Placeholder: Sample reconciliation output]
(e.g., diff in commissions, open position quantities)

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Key Takeaways

- **Performance metrics** must be monitored live with the same rigor as backtest.
- **Cost breakdown** reveals where live trading deviates from assumptions.
- **Attribution analysis** identifies actionable improvements.
- **Kill switch** provides tail-risk protection.
- **Reconciliation** ensures data integrity and correct P&L.

Next: Incorporate live data into updated cost models and refine execution logic.

Thank You

Questions & Discussion

Appendix: Live Performance Metrics — Formulas

All metrics use **daily total return** (including dividends, net of all costs).

$$\text{CAGR} = \left(\frac{\text{NAV}_{\text{end}}}{\text{NAV}_{\text{start}}} \right)^{\frac{252}{T}} - 1 \quad \text{Volatility} = \sqrt{252} \times \sigma_{\text{daily}}$$

$$\text{Sharpe} = \frac{\text{CAGR} - r_f}{\text{Volatility}} \quad \text{Max DD} = \max_{\tau} \left(1 - \frac{\text{NAV}_{\tau}}{\max_{t \leq \tau} \text{NAV}_t} \right)$$

$$\text{Beta} = \frac{\text{Cov}(R_{\text{strategy}}, R_{\text{SPY}})}{\text{Var}(R_{\text{SPY}})} \quad \text{Alpha} = \text{CAGR} - r_f - \beta (R_{\text{SPY}} - r_f)$$

$$\text{Win rate} = \# \text{ winners} / \# \text{ closed trades.} \quad \text{Turnover} = \frac{1}{N} \sum_t |\Delta w_t|.$$